

Explainable Deep Generative Models for Conditional Drug Discovery

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Problem statement

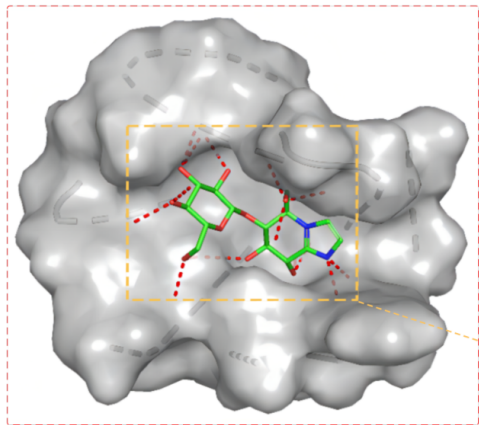


Figure 1: Drug-Protein binding. Adapted from Zhang et al. (2022)

Goal

Given a target protein, generate candidate ligands satisfying:

- Bind strongly to the target
- Are chemically valid
- Are easy to synthesize in a laboratory
- Have confidence estimates on the binding affinity
- Provide an interpretable rationale (i.e., “why is this a good candidate?”)

Deep generative models nowadays are capable of designing new molecules for a given protein target. However, some issues remain:

- Most models generate nonsense drugs (e.g., containing carbon atoms with 6 bonds).
- They are black boxes: no information on how reliable a proposal is or why it was proposed.
- False positives $\approx$ months of wasted wet-lab effort.
- Anyone can make a GPT-style drug generator, but why should we trust it?

“Would you take a drug suggested by your ML model?”

Three areas for improvement

State-of-the-art models focus on maximizing metrics such as predicted affinity and selectivity. Other essential factors are overlooked.

This project's objective is to incorporate the following dimensions:

- Biochemical restrictions.
- Uncertainty estimation.
- Mechanistic explainability.

Case study: disrupt proteins of the *Trypanosoma cruzi* proteome (Chagas disease).

- **O1)** A diffusion-based drug generator conditioned on an input protein binding pocket, with biochemical restrictions encoded in the architecture rather than via auxiliary losses.
- **O2)** Affinity estimates with calibrated uncertainty by approximate Bayesian inference over parameters, and distribution-free conformal layers giving prediction sets.
- **O3)** Plausible mechanistic explanations for why a molecule was suggested via sparse autoencoders and causal tracing.

Extra: a *T. cruzi* case study with partner labs, and an open-source web interface so pharmacologists can query their own targets.

- **Q1)** How do we encode biochemical restrictions ($SE(3)$ -equivariance, valence and chirality constraints, pocket geometry etc) as architectural restrictions instead of loss terms?
- **Q2)** How do we get calibrated uncertainty for predicted activity, and transfer those guarantees to generated candidates under distribution shift?
- **Q3)** Can we find internal components that map to chemical concepts? Do LLM explainability tools transfer to equivariant message-passing networks?

We frame drug discovery as conditional density estimation on a structured probability space. Given a pocket representation $c = \phi(\text{protein})$, learn

$$p_{\theta}(\mathcal{G} \mid c), \quad \mathcal{G} = (V, E), \quad v_i = (x_i, a_i) \in \mathbb{R}^3 \times \mathcal{A},$$

i.e. molecules as graphs: each node (atom) v_i carries a 3D coordinate x_i and other chemical features a_i (atom type, charge, ...), and edges E encode bonds.

The first step is to define precise model classes and a taxonomy of inductive biases for drug-protein binding.

1. Constrained generation

We apply an SE(3)-equivariant denoising diffusion model like EDM (Hoogeboom et al. 2022), DiffSBDD (Schneuing et al. 2024), conditioned on pockets from AlphaFold (Jumper et al. 2021) / ESMFold (Lin et al. 2023).

We enforce two kinds of chemical prior, handled differently:

- **Geometric symmetries**: preserved exactly via equivariant GNNs (Satorras et al. 2021); permutation equivariance from message passing on atom multisets.
- **Chemical constraints** (valence, aromaticity, ...): via classifier-free guidance (Ho et al. 2022) with a validity discriminator, or autoregressive validity checks.

$$\tilde{\epsilon}_{\theta}(x_t | c) = \epsilon_{\theta}(x_t | \emptyset) + w (\epsilon_{\theta}(x_t | c) - \epsilon_{\theta}(x_t | \emptyset))$$

Open question: does this actually beat post-hoc filtering? Goal: valid by construction.

1. Constrained generation

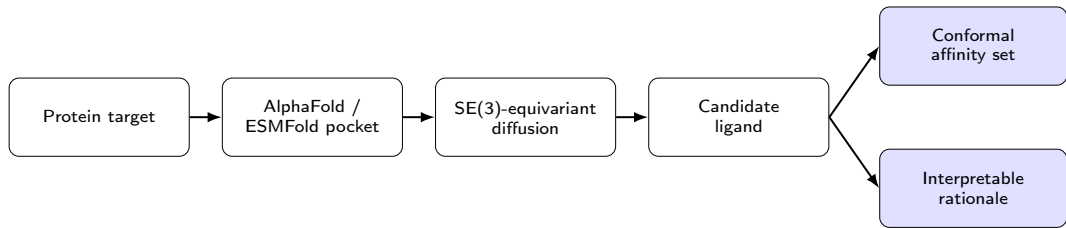


Figure 2: We generate a candidate drug from a target protein, while providing confidence estimates and plausible rationales.

2. Calibrated uncertainty

- **Parameter side:** compare last-layer Laplace (Daxberger et al. 2021) (closed-form predictive), SGLD (Welling et al. 2011), and deep ensembles (Lakshminarayanan et al. 2016). Calibration via expected calibration error and reliability diagrams (Guo et al. 2017).
- **Prediction side:** a separately trained affinity predictor wrapped in a conformal layer (Angelopoulos et al. 2023), producing a prediction set per molecule.

Conformal coverage holds under exchangeability with the calibration set:

$$(Z_1, \dots, Z_n, Z_{n+1}) \stackrel{d}{=} (Z_{\pi(1)}, \dots, Z_{\pi(n)}, Z_{\pi(n+1)}), \quad \forall \pi, \quad Z_i = (X_i, Y_i),$$
$$\Rightarrow \quad \mathbb{P}(Y_{n+1} \in \mathcal{C}_\alpha(X_{n+1})) \geq 1 - \alpha.$$

2. Calibrated uncertainty

However, generated molecules are not exchangeable with the calibration set!
They come from a shifted, model-induced distribution, so standard guarantees break.

Fixes:

- Conditional / group-balanced conformal (Romano et al. 2019; Gibbs et al. 2023).
- Weighted conformal under covariate shift (Tibshirani et al. 2019); coverage gap bounded by the total-variation distance (Barber et al. 2022):

$$\mathbb{P}(Y \in \mathcal{C}_\alpha(X)) \geq 1 - \alpha - \text{TV}(P_{\text{cal}}, P_{\text{gen}}).$$

The two methods cover each other's weaknesses: Bayesian predictive variance becomes the non-conformity score for the conformal layer (Romano et al. 2019). High-confidence molecules should be exactly the ones the model is right about.

3. Mechanistic explainability

So far, explainability has relied on SHAP / LIME heatmaps (Lundberg et al. 2017; Ribeiro et al. 2016). They show which inputs mattered, not logic the model applied. Also, they were shown to be unstable and provably unreliable for common tasks (Adebayo et al. 2018).

Three tools, ported from LLM explainability:

- **Sparse autoencoders**: decompose internal activations into a few interpretable features (Cunningham et al. 2023).
- **Activation patching**: swap activations across runs to find functional components (Vig et al. 2020).
- **Causal tracing**: intervene on representations to measure cause, not correlation (Meng et al. 2022).

Evaluation: intervention-based faithfulness tests (Wang et al. 2022).

Case study: *T. cruzi*

Chagas disease is a neglected tropical disease with unsatisfactory chronic-stage treatment.

- Partner labs prioritise a subset of the *T. cruzi* proteome.
- For each target, our pipeline returns candidate ligands, a conformational affinity set, and an interpretable rationale.
- Survivors of preliminary screening progress to wet-lab binding and parasiticide assays (where possible).
- Experimental data, when obtained, act as a test of the calibration and interpretability claims.

From the ML side:

- Constrained denoising as a principled way to enforce hard priors in diffusion models.
- Conformal layers inside generative pipelines, with explicit covariate-shift accounting.
- Mechanistic interpretability adapted to architectures for structured molecular data.

From the bioinformatics side:

- An open-source generator giving a calibrated affinity set and a rationale per candidate.
- Prioritised, novel candidates for wet-lab validation.
- Giving pharmacologists testable hypotheses and means of flagging hallucinations.

- **Università degli Studi di Milano:** Dipartimento di Informatica, AnacletoLab. Aligned with Profs. Giorgio Valentini and Elena Casiraghi (ML, bioinformatics, AI for biomedicine).
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The methods also transfer to RNA, peptide, and lipid generation.

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